arranged tiers and rows of cells. It is broken here and there by the presence of lenticels.

Phelloids

Phellem (Cork) like cells which lack suberin in their walls.

Phellogen (Cork Cambium)

It is a secondary lateral meristem. It comprises homogenous meristematic cells unlike vascular cambium. It arises from epidermis, cortex, phloem or pericycle (extrastelar in origin). Its cells divide periclinally and produce radially arranged files of cells. The cells towards the outer side differentiate into phellem (cork) and those towards the inside as phelloderm (secondary cortex).

Phelloderm (Secondary cortex)

It is a tissue resembling cortical living parenchyma produced centripetally (inward) from the phellogen as a part of the periderm of stems and roots in seed plants.

Differences Between Phellem and Phelloderm

Phellem (Cork)	Phelloderm (Secondary cortex)
It is formed on the outer side of phellogen.	It is formed on the inner side of phellogen.
Cells are compactly arranged in regular tires and rows without intercellular spaces.	Cells are loosely arranged with intercellular spaces.
Protective in function.	As it contains chloroplast, it synthesises and stores food.
Consists of non-living cells with suberized walls.	Consists of living cells, parenchymatous in nature and does not have suberin.
Lenticels are present.	Lenticels are absent.

DO YOU KNOW?

Rhytidome is a technical term used for the outer dead bark which consists of periderm and isolated cortical or phloem tissues formed during successive secondary growth. Example: *Quercus*.

Polyderm is found in the roots and underground stems.eg. Rosaceae. It refers to a special type of protective tissues consisting of uniseriate suberized layer alternating with multiseriate nonsuberized cells in periderm.

Bark

The term 'bark' is commonly applied to all the tissues outside the vascular cambium of stem (**i.e., periderm, cortex, primary phloem and secondary phloem**). Bark protects the plant from parasitic fungi and insects, prevents water loss by evaporation and guards against variations of external temperature. It is an insect repellent, decay proof, fireproof and is used in obtaining drugs or spices. The phloem cells of the bark are involved in conduction of food while secondary cortical cells involved in storage.

If the phellogen forms a complete cylinder around the stem, it gives rise to **ring barks**. Example: *Quercus*. When the bark is formed in overlapping scale like layers, it is known as **scale bark**. Example: Guava. While ring barks normally do not peeled off, scale barks peeled off.



Figure 10.12: *Quercus* Tree-showing ring bark



Figure 10.13: Guava tree showing scale bark

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